Practice Problem Sheet – 7
Transform Calculus
(MA20202)

Koeli Ghoshal

These answers are based on $F_c[f(x)] = \int_0^\infty f(x)\cos\alpha x\,dx$, $f(x) = \frac{2}{\pi}\int_0^\infty F_c\,d\alpha$ and similar formulae.

Q1. Find the complex form of the Fourier integral representation for the function

$$f(x) = \begin{cases} |x| & -\pi < x < \pi \\ 0 & \text{otherwise} \end{cases}$$

Ans: $f(x) = \frac{1}{\pi}\int_{-\infty}^{\infty} \frac{1}{\alpha^2}\left[\cos\pi\alpha - \pi\alpha\sin\pi\alpha - 1\right] e^{i\alpha x}\,d\alpha$

Q2. Find the Fourier transform of

$$f(x) = \begin{cases} 1-x^2 & |x| \le 1 \\ 0 & |x| > 1 \end{cases}$$

and hence evaluate $\int_0^\infty \left(\frac{x\cos x - \sin x}{x^3}\right)\cos\frac{x}{2}\,dx$

Ans: $F_c[f(x)] = 2\,\frac{\sin\alpha - \alpha\cos\alpha}{\alpha^3}$ [even f^n, so F.T. ≃ F.C.T]

Given integral $= -\frac{3\pi}{16}$

Q3. Find the F.C.T. of $f(x) = \begin{cases} \cos x & 0 < x < a \\ 0 & x > a \end{cases}$

Ans: $F_c[f(x)] = \frac{1}{2}\left[\frac{\sin(\alpha+1)a}{\alpha+1} + \frac{\sin(\alpha-1)a}{\alpha-1}\right]$

Q4. Using inverse sine transform, find $f(x)$ if

$$F_s(\alpha) = \frac{1}{\alpha}e^{-a\alpha}, \quad a > 0$$

Ans: $f(x) = \frac{2}{\pi}\tan^{-1}\frac{x}{a}$

Q5. Find $f(x)$ if $\int_0^\infty f(x)\sin\alpha x\,dx = \begin{cases} 1 & 0 \le \alpha < 1 \\ 2 & 1 \le \alpha < 2 \\ 0 & \alpha \ge 2 \end{cases}$

Ans: $f(x) = \frac{2}{\pi}\left(\frac{1+\cos x - 2\cos 2x}{x}\right)$

Q6. Find the Fourier transform of the function $e^{-ax} u_0(x)$, where $u_0(x)$ is the unit step function

or find the F.T. of $f(x) = \begin{cases} e^{-ax}, & x \geq 0 \\ 0 & x < 0 \end{cases}$ $a > 0$

Ans: $F\left[e^{-ax} u_0(x)\right] = \frac{1}{a + i\alpha}$

Q7. Find $F^{-1}\left[\frac{e^{4i\alpha}}{3 + i\alpha}\right]$

Ans: Use result from previous problem and shifting property

answer is $e^{-3(x+4)} u_{-4}(x) = \begin{cases} 0 & x < -4 \\ e^{-3(x+4)} & x \geq -4 \end{cases}$

Q8. Given $f(x) = 4e^{-|x|} - 5e^{-3|x+2|}$

evaluate its F. transform $F(\alpha)$.

Ans: $F(\alpha) = \frac{8}{\alpha^2 + 1} - \frac{30 e^{2i\alpha}}{\alpha^2 + 9}$

Q9. Find $F^{-1}\left[F(\alpha)\right]$ where $F(\alpha) = e^{-2|\alpha|}$

Ans: $\frac{2}{\pi} \frac{1}{x^2 + 4}$

Q10. Evaluate $F^{-1}\left\{\frac{1}{\alpha^2 + 4\alpha + 13}\right\}$

Ans: $e^{-2ix}\left[\frac{1}{6} e^{-3|x|}\right]$

Q11. Without using the definition, find the Fourier cosine and sine transform of $f(x) = e^{-ax}$ $x \geq 0$ $a > 0$.

Ans: Use the formula for derivatives i.e. $F_c[f''(x)]$ and $F_s[f''(x)]$. $F_c(\alpha) = \frac{a}{a^2 + \alpha^2}$, $F_s(\alpha) = \frac{\alpha}{a^2 + \alpha^2}$

**** The End ****